
```

clc;
clear;
close all;

% Parameters
Am = 1;           % Message amplitude (V)
fm = 100;         % Message frequency (Hz)
Ac = 5;           % Carrier amplitude (V)
fc = 1000;        % Carrier frequency (Hz)
fs = 20*fc;       % Sampling frequency (Hz)
kp = pi/2;        % Phase sensitivity (rad/V)

% Time Vector
t = 0:1/fs:4/fm;

% Message Signal
m = Am*sin(2*pi*fm*t);

% Phase Modulated Signal
pm = Ac*cos(2*pi*fc*t + kp*m);

% Phase Deviation / PM Modulation Index
beta = kp*Am;

% FFT
N = length(pm);
f = (-N/2:N/2-1)*fs/N;
PM_FFT = abs(fftshift(fft(pm)))/N;

% Power Spectral Density
PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);

% Approximate Bandwidth using Carson's Rule
BW = 2*(beta+1)*fm;

% Display Results
fprintf('Phase Deviation = %.2f radians\n', beta);
fprintf('PM Modulation Index = %.2f\n', beta);
fprintf('Estimated Bandwidth = %.2f Hz\n', BW);

% Plotting
figure('Name', 'PSD Analysis of PM Signal', 'NumberTitle', 'off');

subplot(2,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(2,2,2);
plot(t,pm);

```

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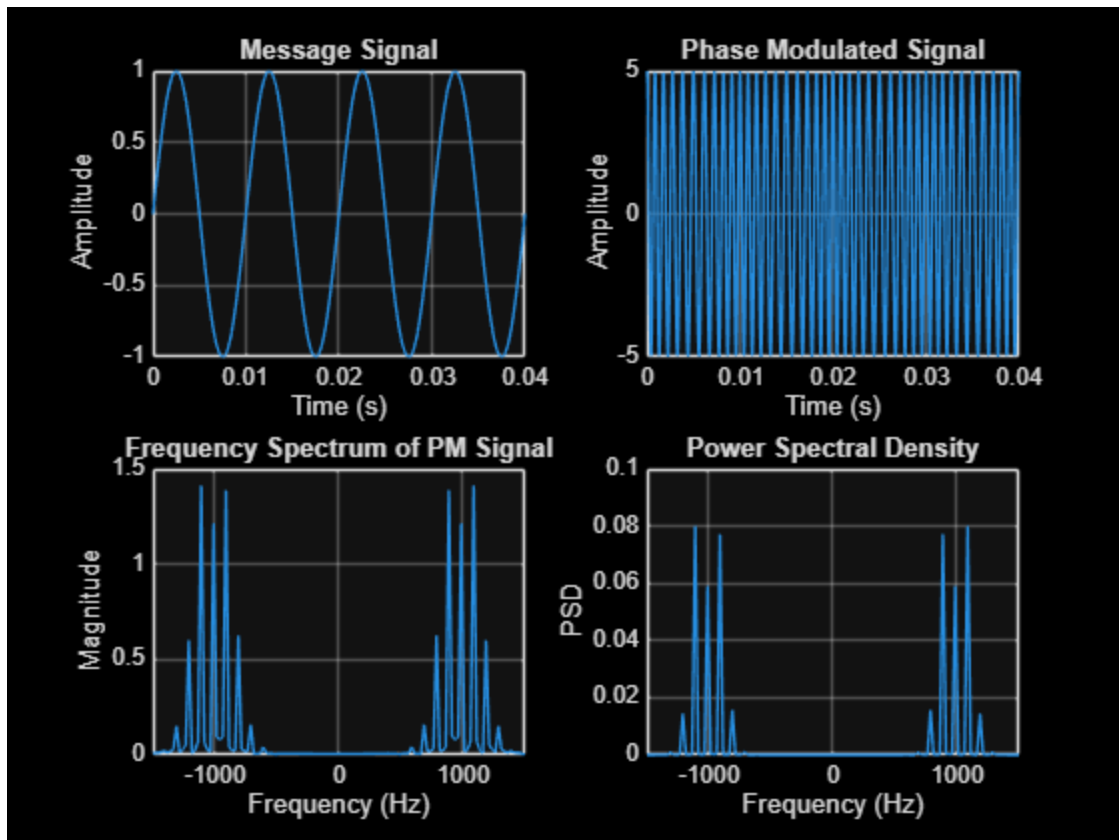
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(2,2,3);
plot(f,PM_FFT);
xlim([-1500 1500]);
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

subplot(2,2,4);
plot(f,PSD);
xlim([-1500 1500]);
title('Power Spectral Density');
xlabel('Frequency (Hz)');
ylabel('PSD');
grid on;

```

Phase Deviation = 1.57 radians
PM Modulation Index = 1.57
Estimated Bandwidth = 514.16 Hz



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